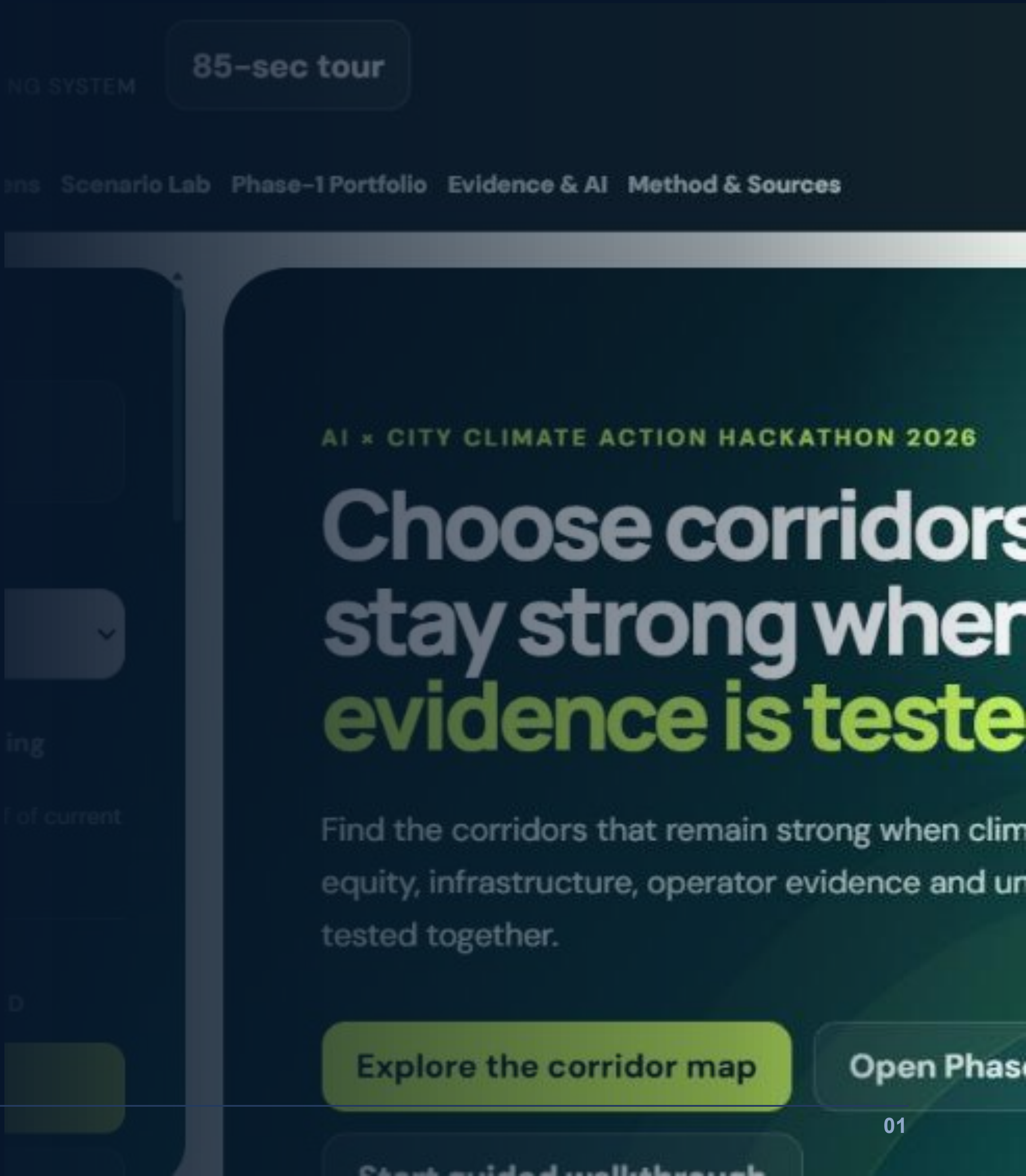
GitHub: https://github.com/qjmre23/Route2Zero | Live: https://route2zero.netlify.app/

ROUTE2ZERO

AI/ML-assisted planning for a just e-jeepney transition.

Which Metro Manila corridors should be validated and prioritized first — and how robust is that decision?



1,522 route-direction records screened

9 robust-priority corridors

0 current route validations

Team Larpers · Metro Manila · AI x City Climate Action Hackathon 2026

WHY THIS MATTERS

Transport electrification is a climate opportunity — prioritization is the bottleneck.

34%

of Philippine GHG emissions came from transport in 2015

80%

of transport emissions were attributed to road transport

NATIONAL DIRECTION

50% EV adoption by 2040

A roadmap target — not an achieved value and not route-level proof.

Cities still must decide where limited validation, planning, charging, and transition support create the strongest public value first.

2025 DOE direction: distribution utilities integrate EV charging demand into development planning.

Sources: DOE CREVI (2023); DOE DC2025-08-0012 and official 2025 releases.

THE PLANNING GAP

The evidence for one corridor decision lives in different systems.

Route + service

historic schedules, current status

Climate + energy

vehicle, grid, and operating assumptions

Equity + access

population and validated area evidence

Charging

mapped proximity versus verified capacity

Operators

fleet, depot, finance, governance

Uncertainty

data quality and policy sensitivity

Evidence does not agree by default.

High activity can still mean weak charging evidence.
High equity can rest on outdated service records.
A top score can move when policy priorities change.

Route2Zero integrates the evidence — it does not erase disagreement.

DECISION FRAME

Route2Zero answers one city decision.

Which corridors should Metro Manila validate and prioritize for e-jeepney transition — and how robust is that recommendation?

- **Where is the climate value?**
- **Who benefits — and what evidence supports that?**
- **Can charging and operators support the transition?**
- **Does the recommendation survive changing assumptions?**

Output: priority · evidence grade · climate range · stability · portfolio status · next validation action

USERS AND BENEFICIARIES

Built for the institutions that must make the transition work.

Beneficiaries: transit-dependent riders, drivers and operators navigating modernization, communities along high-value corridors, and public agencies allocating limited assessment capacity.

PRODUCT PROMISE

From a leaderboard to a city electrification decision laboratory.

01	02	03	04	05	06
Screen	**Estimate**	**Quantify**	**Test**	**Stress-test**	**Assemble**
1,522 route directions	missing service intelligence	climate + energy scenarios	equity, charging, operators	rank stability + evidence	constrained Phase-1 portfolio

Planning & Evidence Assistant

Explains precomputed evidence, compares scenarios, and turns missing evidence into a fieldwork queue.

Humans decide

Policy weights and constraints remain explicit city choices.

AI / ML BOUNDARY

AI has analytical work — policy authority stays human.

MACHINE LEARNING

Service-intensity estimate
Anomaly flag
Corridor typology

DETERMINISTIC MODELS

Climate + energy
Evidence confidence
Sensitivity + optimization

LLM PLANNING ASSISTANT

Explain evidence
Compare scenarios
Triage validation

ML estimates. Deterministic models quantify. City teams control policy. The LLM explains and triages evidence.

The assistant never writes scores, rankings, climate values, or weights.

DATA, PROVENANCE, VALIDATION

Every recommendation carries an evidence trail.

SOURCE REGISTRY

- Historic Sakay GTFS
- WorldPop 2020
- DOE energy context
- OSM infrastructure snapshot
- Operator neutral prior
- Current validation ledgers

CLAIM STATUS

OBSERVED

DERIVED

ML_ESTIMATED

PROXY

SCENARIO

NEUTRAL PRIOR

MISSING

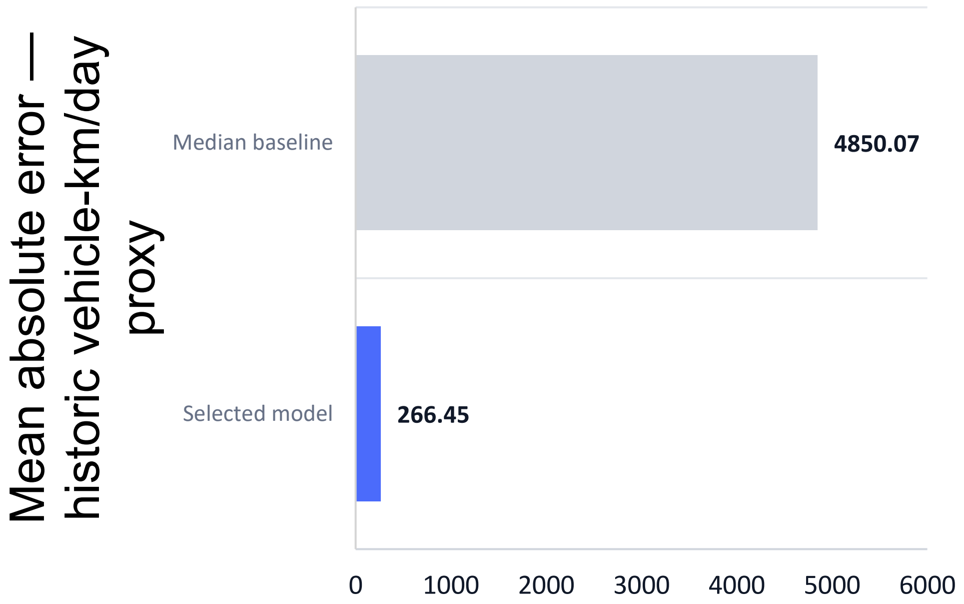
1,522
screening records

0
current validations

6
registered source families

ML SERVICE INTELLIGENCE

Machine learning estimates service intensity where evidence is incomplete.



HistGradientBoostingRegressor

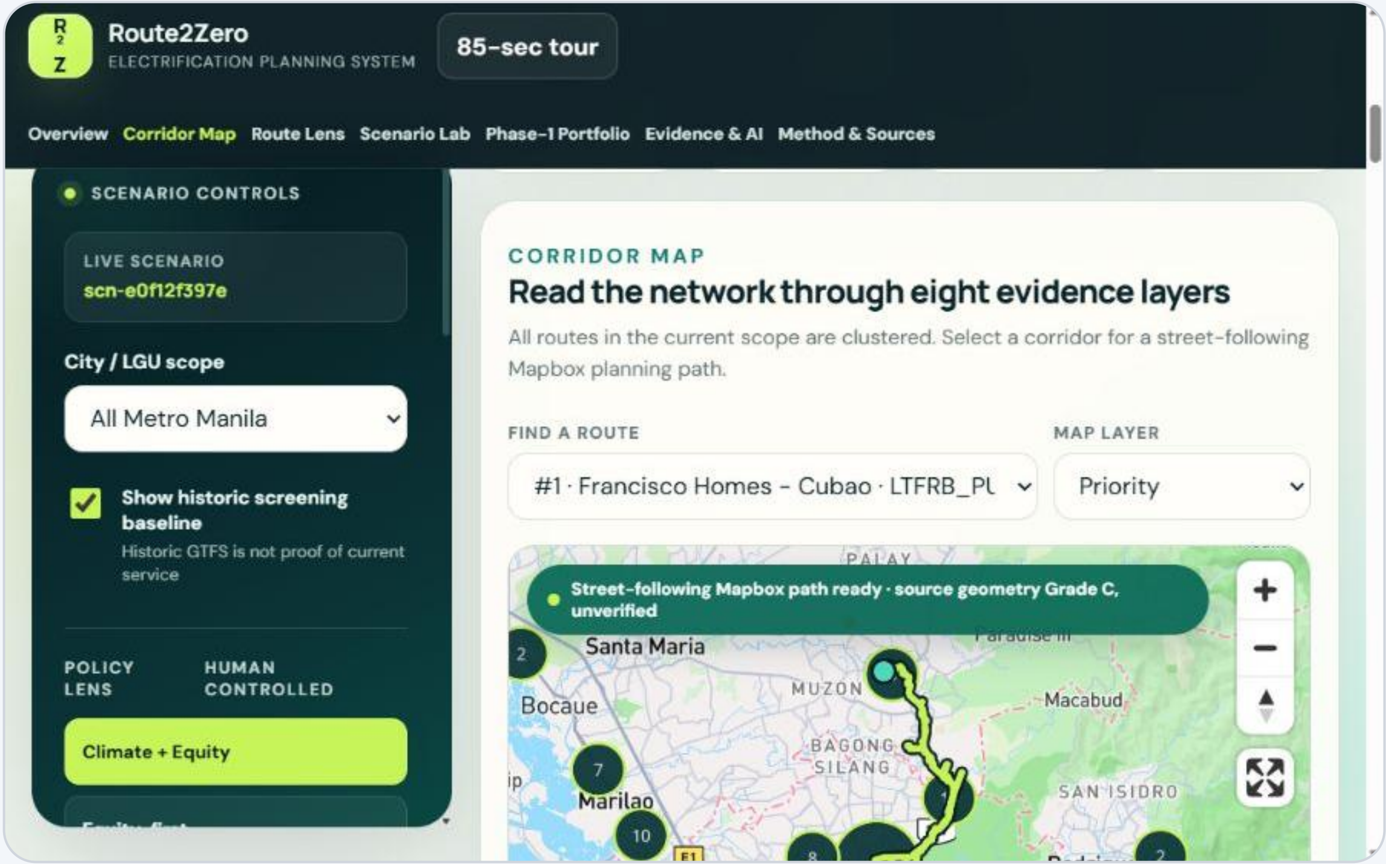
TARGET	Historic schedule-based vehicle-km/day proxy
VALIDATION	GroupKFold(5) by normalized corridor
ROWS	1,521 training records · 714 corridor groups
METRICS	MAE 266.45 · RMSE 562.80 · R^2 0.9911
VERSION	service-v1-266e0b1d

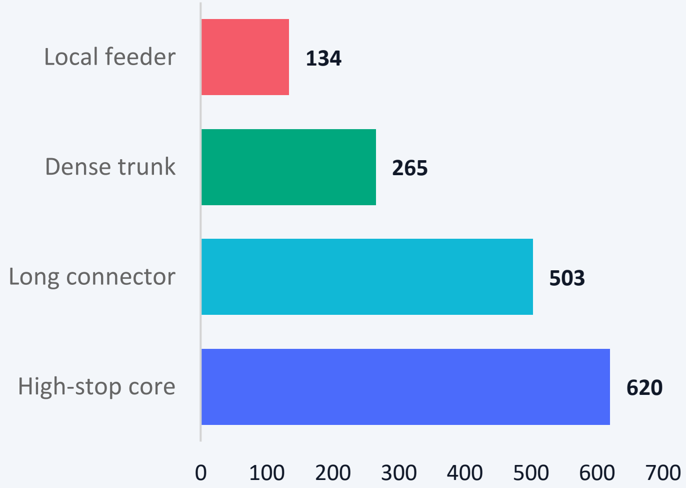
Flagship: historic proxy 31,585 vehicle-km/day; ML estimate 29,812. Historic evidence is used. ML is comparison-only and does not mean ridership.

Source: model_metrics.json · grouped corridor validation · no score/rank leakage features

CORRIDOR TYPOLOGY

Route2Zero compares like with like.





SELECTED K = 4

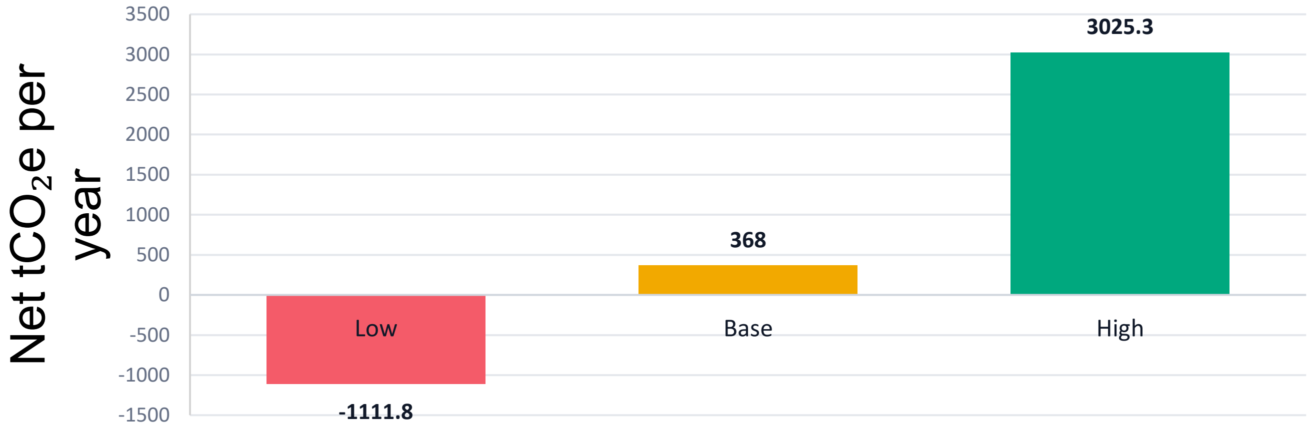
Silhouette 0.270

KMeans · typology-v1-f9da2ebe
Typology informs comparison, not policy score.

CLIMATE + ENERGY IMPACT

The system estimates climate outcomes — not an ‘emissions score.’

31,585 daily VKT → **30–70%** electric share → **11.1–13.5** MWh/day → **−1,112 to +3,025** tCO_2e/year



SCENARIO — not measured reduction

Grid: 0.718 / 0.55 / 0.35 $kgCO_2e/kWh$
EV use: 1.05 / 0.75 / 0.55 kWh/km
300 operating days/year
Low case can increase emissions

Flagship: Francisco Homes–Cubao · climate-v1.0 · DOE context + explicit scenario assumptions

EQUITY + ACCESSIBILITY

Climate value is not enough if access is left behind.

77.42
population-exposure score

25 / 100
equity evidence confidence

PROXY
claim status

What exists

WorldPop population exposure
Route-level spatial overlap
Transparent missing-data policy

What does not yet exist

Validated socioeconomic layer
Accessibility-gap or transit-dependence layer
Settlement-status evidence

Low evidence is not low need. Population density is never used to infer poverty or informal-settlement status.

CHARGING + OPERATOR READINESS

A climate opportunity is not implementation-ready without infrastructure and operators.

CHARGING EVIDENCE

60.45

readiness score · PROXY

Energy need	13.16 MWh/day · base
Nearest mapped substation	0.56 km
Nearest mapped charger	0.65 km
Candidate terminals	2 · unverified

OPERATOR EVIDENCE

50.0

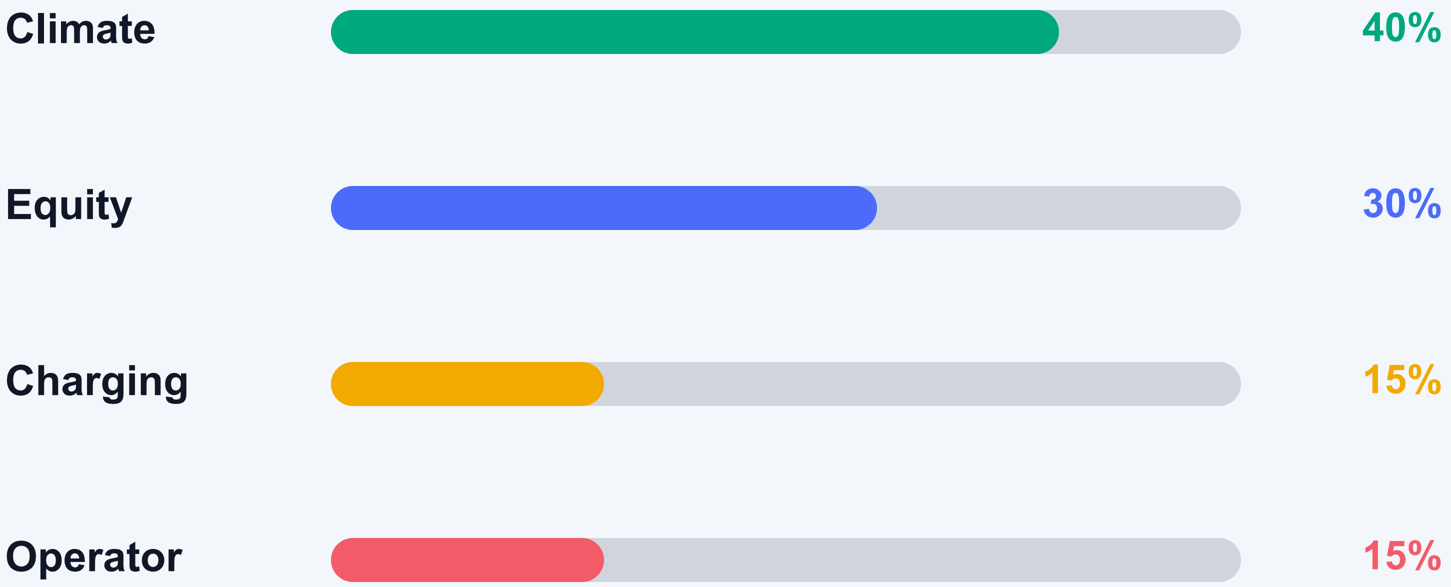
neutral policy prior

Observed operator score	NOT AVAILABLE
Evidence completeness	0%
Evidence confidence	5 / 100
Claim status	NEUTRAL PRIOR

UTILITY CAPACITY: NOT VERIFIED — mapped proximity is not capacity, site control, or interconnection approval.

HUMAN-CONTROLLED POLICY MODEL

The policy trade-offs stay visible.



DEFAULT SCENARIO

Climate + Equity

scn-e0f12f397e

Named alternatives

Equity-first: 25 / 50 / 15 / 10
Infrastructure-first: 25 / 20 / 35 / 20

Weights are normalized to 100%. They are a city choice — never presented as objectively correct.

UNCERTAINTY + RANK STABILITY

A recommendation is stronger when it survives changing assumptions.

5,000

fixed-seed policy-weight simulations

Francisco Homes–Cubao

rank 1

100% top-10

ROBUST PRIORITY

P10 1 · P90 1

Binangonan–JRC via Angono

rank 11

40.7% top-10

SCENARIO-DEPENDENT

P10 9 · P90 24

Rank stability measures resilience to tested policy weights — not model accuracy.

VALUE OF INFORMATION

Route2Zero tells the city which missing evidence is worth collecting first.

Climate assumptions	**rank swing up to 1,395**	Portfolio flip possible
Equity population exposure	**rank swing up to 7**	Validate area evidence
Charging readiness	**rank swing up to 3**	Request utility/site evidence
Operator readiness	**rank swing up to 3**	Interview operator/cooperative

Next action: calibrate vehicle efficiency, electrification share, and grid assumptions before treating the flagship climate case as decision-ready.

CONSTRAINED PHASE-1 PORTFOLIO

Cities fund programs, not leaderboards.

SIMPLE TOP 8		OPTIMIZED 8
PUJ1352 removed		PUJ1638 added
PUJ1240 removed	→	PUJ1153 added
PUJ2084 removed		PUJ1350 added
PUJ1157 removed		PUJ1405 added

MAX 8 | MAX 2 / CITY | MAX 1 DIRECTION / CORRIDOR | EQUITY ≥ 40 | EVIDENCE ≥ C

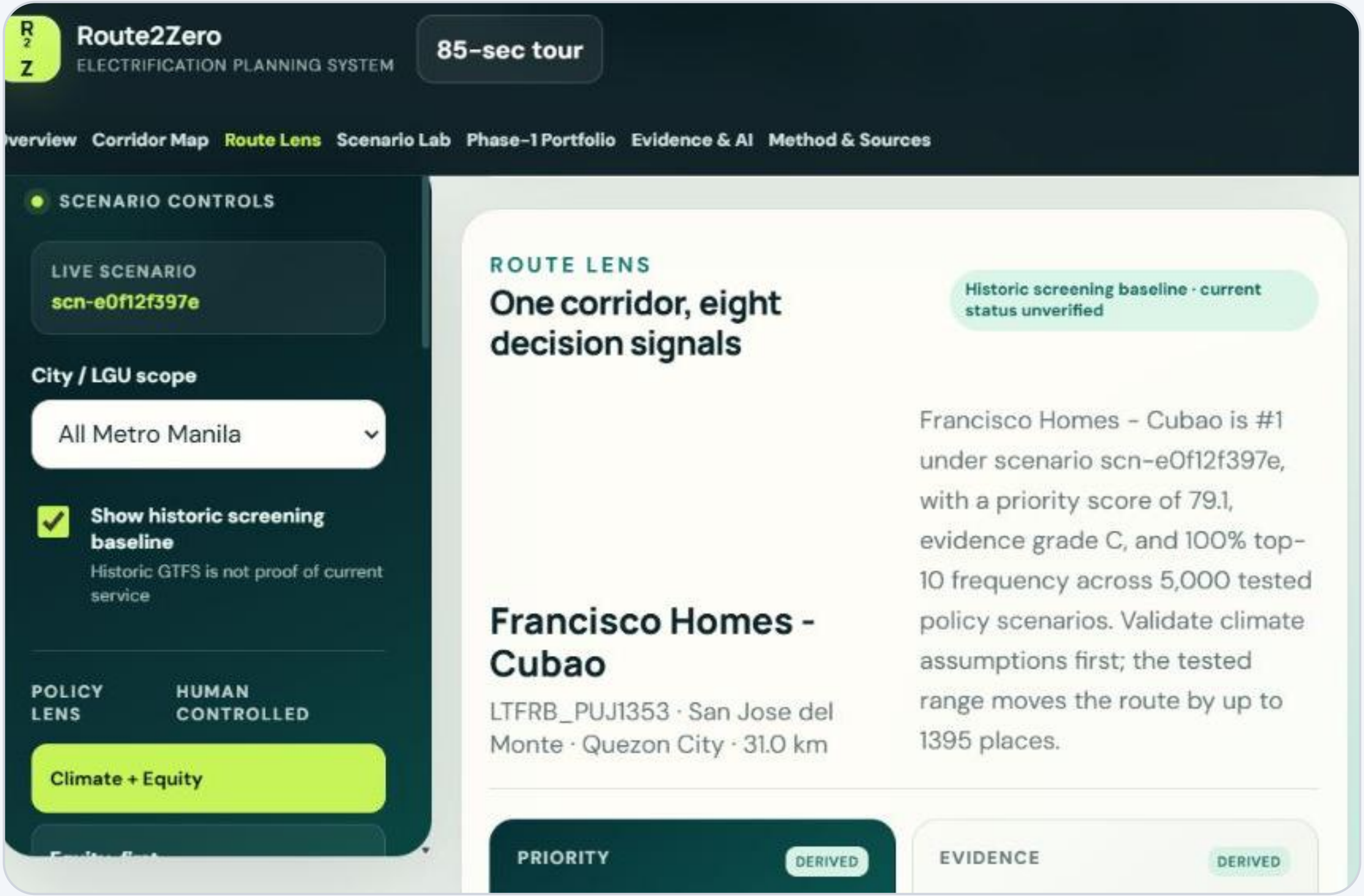
Portfolio scenario range

−8,191 to +22,288 tCO_2e/year

Avg equity 73.9 · all 8 grade C · no hypothetical budget

FLAGSHIP · FRANCISCO HOMES–CUBAO

One corridor, from raw evidence to pilot decision.



RANK / PRIORITY	1 / 79.07
EVIDENCE	C · 38.34
STABILITY	100% top-10
CLIMATE	−1,112 to +3,025 tCO_2e/y
CHARGING	60.45 · capacity unverified
OPERATOR	50 neutral prior
TYPOLOGY	Dense Urban Trunk · outlier
PORTFOLIO	Selected

Proceed to pilot validation — not procurement.

PILOT PATHWAY

The next six months convert assumptions into city-owned evidence.

- **M1** — Baseline + protocol
- **M2** — Refresh + ground truth
- **M3** — Model + route validation
- **M4** — Climate / grid / operator
- **M5** — Scenario co-design
- **M6** — Decision pack + handover

GATE 1
Evidence plan + protocol

GATE 2
Route set + quality review

GATE 3
Decision pack + reproducibility

Participants LGU · operator/cooperative · utility/energy partner · riders/community · Team Larpers

Outcome: a validated, reproducible shortlist with named owners for every unresolved evidence gap.

IMPACT · DIFFERENTIATION · SCALE

From screening 1,522 routes to making a defensible first move.

EXPECTED IMPACT

8 corridors advance to deeper validation
Portfolio range: −8.2k to +22.3k tCO_2e/year
Evidence grade improvement is a pilot target

WHY ROUTE2ZERO

Not one black-box score
ML + climate + GIS + uncertainty + portfolio
Humans control policy and constraints

HOW IT SCALES

Metro Manila is the deep pilot
Adapters for routes, population, boundaries, energy, operators
Every new city requires recalibration

Pilot Route2Zero with city, operator, and energy partners to turn an evidence-aware prototype into a validated transition-planning workflow.

PREDICT → QUANTIFY → STRESS-TEST → VALIDATE → ACT

Team Larpers · John Marwin Ebona · Isaac Marcus · Andrei Dela Cruz · Russel Mendez
TRISTIAN JAMES CABALAR · JOHN MICHAEL PALAGANAS · Carl Nueva · JOSEPH CLARENCE PARAYAOAN